

Formula Explanation

How VLOOKUP, INDEX-MATCH, and nested IF formulas power the dashboard

Financial_Health_Dashboard.xlsx contains four linked sheets: **Raw Data** (source financials), **Benchmarks** (rating thresholds), **Ratio Calculations** (the 14 ratio formulas), and **Dashboard** (KPI cards and charts). No ratio value is typed in directly — every number is pulled and calculated live from the Raw Data sheet, so changing a single input automatically recalculates every ratio, rating, and chart in the workbook.

1. Two-Way INDEX-MATCH: pulling values by line item and year

Each ratio needs a specific line item (e.g. “Total Current Assets”) for a specific year (e.g. 2024). Rather than pointing to a fixed cell — which would break if a row were inserted or the year columns were reordered — every lookup uses a two-way INDEX-MATCH: one MATCH finds the row for the line item, a second MATCH finds the column for the year, and INDEX returns the value at their intersection.

```
=INDEX('Raw Data'!$B$5:$F$40,  
MATCH("Total Current Assets", 'Raw Data'!$A$5:$A$40, 0),  
MATCH(C$4, 'Raw Data'!$B$4:$F$4, 0))
```

The Current Ratio for 2024, for example, wraps two of these lookups together:

```
=INDEX('Raw Data'!$B$5:$F$40,  
MATCH("Total Current Assets", 'Raw Data'!$A$5:$A$40, 0),  
MATCH(G$4, 'Raw Data'!$B$4:$F$4, 0))  
/  
INDEX('Raw Data'!$B$5:$F$40,  
MATCH("Total Current Liabilities", 'Raw Data'!$A$5:$A$40, 0),  
MATCH(G$4, 'Raw Data'!$B$4:$F$4, 0))
```

This pattern is used for all 14 ratios across all 5 years — 70 individual INDEX-MATCH lookups in total on the Ratio Calculations sheet.

2. VLOOKUP: pulling benchmark thresholds

The Benchmarks sheet holds a reference table of every ratio's name, its favorable direction (Higher or Lower), and its Strong / Moderate threshold values. The Ratio Calculations sheet looks up each ratio's row by name with VLOOKUP so the benchmark table only has to be maintained in one place — update a threshold once and every dependent formula picks it up automatically.

```
Direction: =VLOOKUP($B5, Benchmarks!$A$5:$E$18, 2, FALSE)  
Strong ≥/≤: =VLOOKUP($B5, Benchmarks!$A$5:$E$18, 3, FALSE)  
Moderate: =VLOOKUP($B5, Benchmarks!$A$5:$E$18, 4, FALSE)
```

The FALSE argument forces an exact match on the ratio name, so a typo or missing benchmark row returns a clear #N/A rather than a silently wrong threshold.

3. Nested IF: turning numbers into a Strong / Moderate / Weak rating

Some ratios are healthiest when higher (Current Ratio, ROE) and some are healthiest when lower (Debt-to-Equity, Debt-to-Assets). The rating formula branches on the Direction value pulled by VLOOKUP above, then nests a second IF to separate Strong, Moderate, and Weak:

```
=IF(H5="Higher",  
  IF(G5>=J5, "Strong", IF(G5>=VLOOKUP($B5,Benchmarks!$A$5:$E$18,4,FALSE),  
    "Moderate", "Weak")),  
  IF(G5<=J5, "Strong", IF(G5<=VLOOKUP($B5,Benchmarks!$A$5:$E$18,4,FALSE),  
    "Moderate", "Weak")))
```

Read from the outside in: the outer IF checks whether a higher value is favorable for this ratio. Whichever branch applies, an inner IF first checks against the Strong threshold, and — only if that test fails — a second inner IF checks the Moderate threshold, defaulting to “Weak” if neither is met. Conditional formatting on the Ratio Calculations sheet then colors each rating green, amber, or red automatically based on this text value.

4. Supporting formulas on the Raw Data sheet

Subtotals and derived line items (Gross Profit, Operating Income, Total Assets, Net Income, etc.) are themselves formulas rather than typed values, so the statements stay internally consistent if any input line is changed:

```
Gross Profit =Revenue – Cost of Goods Sold  
Operating Income =Gross Profit – Operating Expenses  
Total Current Assets =SUM(Cash:Other Current Assets)  
Total Liabilities =Total Current Liabilities + Long-Term Debt + Other LT Liabilities  
Total Equity =Total Assets – Total Liabilities
```

5. Dashboard KPI cards

The Dashboard sheet's KPI cards pull the latest-year value and rating straight from the Ratio Calculations sheet with a simple INDEX reference, so the summary view never falls out of sync with the detailed calculations underneath it:

```
=INDEX('Ratio Calculations'!$G$5:$G$5,1) → latest-year ratio value  
=INDEX('Ratio Calculations'!$J$5:$J$5,1) → latest-year rating
```

Tip: In Excel, select any formula cell and press Ctrl+` (or Formulas → Show Formulas) to see every formula on a sheet at once, or click a cell and use Formulas → Evaluate Formula to step through a nested calculation like the ones above one piece at a time.